

Lab 11.5 – Test Suite Summary Document

Course: PRM392 - Mobile Application Development (Flutter)

Lab Module: Lab 11 — Testing & Debugging in Flutter (Taskly App)

Status: Completed & Passed

✓ 9 / 9 Scenarios Passed (100% Success)

1. Test Execution Matrix

Test Type	Target File	Scenarios Tested	Status
Unit	task_model_test.dart	Default values, status switches via toggle()	PASS
Unit	task_repository_test.dart	Adding, deleting, and updating tasks in the repository	PASS
Widget UI	task_list_widget_test.dart	Empty list state displays "No tasks yet. Add one!", text insertion tap, and rendering multiple tasks	PASS
Navigation	task_navigation_test.dart	Tap task card navigates to TaskDetailScreen, loads detailed text field, checks page header title	PASS
Integration	task_integration_test.dart	End-to-End flow: create task "Original title" → open details	PASS

Test Type	Target File	Scenarios Tested	Status
		→ modify to "Updated title" → save → verify list updates	

2. Detailed Behaviors Validated

Unit Level

- **Task Instantiation:** Confirmed newly created tasks default `isCompleted` to `false`.
- **Model Immutability:** Checked that calling `toggle()` returns a new `Task` instance with the inverted completion status, preventing unexpected side-effects.
- **Repository Operations:** Validated standard CRUD operations (adding, updating properties, and deleting using unique task IDs).

Widget & UI Level

- **Empty Layout View:** Verified that when the task list size is zero, the screen renders an empty placeholder text: "No tasks yet. Add one!".
- **Task Creation Dialog Flow:** Simulated entering text in the task bar, tapping the "+" button, pumping the frames, and verifying the item tile is added to the screen.
- **Multi-item Rendering:** Confirmed the screen correctly indexes and renders multiple list items.

Navigation & Integration Level

- **Transitions:** Tap events on `ListTile` widgets navigate successfully to `TaskDetailScreen`. Checked the app bar title "Task Detail" and the text field key `detailTitleField`.
- **End-to-End Lifecycle:** Verified that creating a task, editing its content via the detail editor screen, and saving successfully updates the item on the main dashboard list view.

3. Known Limitations

- **Volatility:** Task states are maintained in-memory. Relaunching the application wipes the list database because persistent storage integration was skipped for this testing lab.
- **Mock Client Environment:** Integration tests are executed on mock clocks. Real network constraints (e.g. timeout, connection drops) are not covered.

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